Asserted Zeros: When a Valid Guarantee Certifies an Unmeasured Quantity

Zayyan Ahmed
School of Electrical Engineering and Computer Science
National University of Sciences and Technology (NUST), Islamabad, Pakistan
zayyan@rainsfall.dev

Abstract

A distribution-free guarantee certifies whatever score it is handed, including one that was never measured. We call this a *measurement-integrity failure*, and document one in a pipeline whose two LLM judges grade spoken answers. It stored option probabilities to five decimals, so anything below 5×10^{-6} became exactly 0.0, and rescaling what survived to sum to one pushed the one remaining option (usually the correct one) to 1.0. A record's score is one minus the true answer's stored probability, so on 90.6% of graded records that score was exactly 0.0: a certainty no judge expressed. Those scores became the held-out set that fixes the release threshold, as conformal prediction does. You sort them, take the one sitting at a position your error budget picks, and release an option whose score falls at or below it. With nine in ten at the manufactured 0.0, that position lands among them, so the threshold is 0.0 and the gate admits exactly the options a judge rounded up to 1.0. It now turns on whether the two judges rounded up to the same option, not on the evidence beneath them. A pre-registered probe found every grading decision we checked unchanged. On the recordings this system actually ran, the gate released nothing at all, with or without the defect. Re-imposed at that resolution on correctly measured scores, the rounding makes the gate release more than measurement does in 36 cases of 42 and fewer in none, certifying answers honest measurement would have withheld. The repair is dull: measure every option or refuse. Five questions about how probabilities are recorded catch this before calibration.

1 Introduction

Verification systems compose formal guarantees over quantities a model produces. The guarantee is proved about a random variable; the system computes something it takes to be that variable; and the composition is sound only if the two are the same. Verification checks the guarantee against the score it was handed; nothing validates that the score is the quantity intended. We show that they can differ: a guarantee can remain mathematically valid while certifying a quantity that was never observed. We call this a *measurement-integrity failure*: the downstream guarantee remains valid, but the quantity supplied to it is not the quantity the system intended to measure. The category is broad: a verifier fed a truncated context measures every probability correctly and still supplies the wrong conditional, so we say which member we demonstrate and which our diagnostic catches: a pipeline *recording a value it never observed*, because its record format has no way to represent *unmeasured*.

The setting is the one this workshop asks about: agentic systems delegate consequential decisions to LLM verifiers, and a natural way to make that verification *reliable* is conformal prediction: calibrate a nonconformity score (ours is $1 - p$ for an option, so a small score means a confident grader) on held-out examples, release only when the calibrated uncertainty set is decisive [Vovk et al., 2005,

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Who Verifies the Agents? Toward Reliable Agent Development.

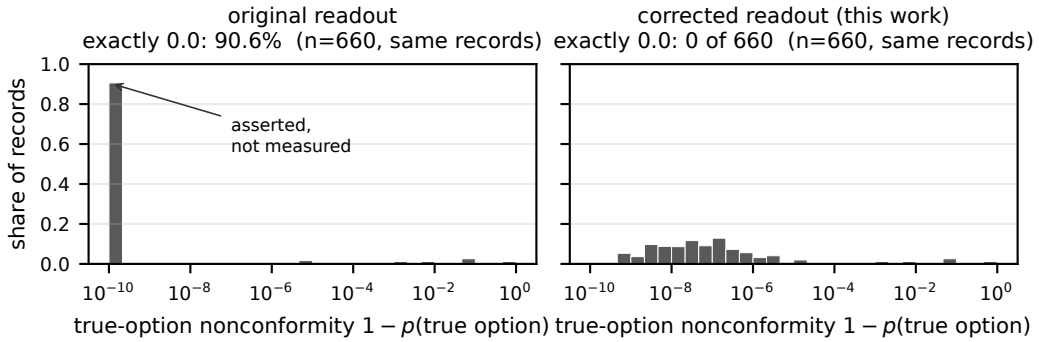


Figure 1: True-option nonconformity on the same 660 graded records, before and after. Left: the original readout, whose spike at exactly 0.0 is asserted, not measured. Right: the same records re-read by the corrected readout of §3, same models, same prompt; 0 of 660 scores are exactly zero (minimum 4.1×10^{-10}). Axes are shared, and exact zeros are drawn at the 10^{-10} axis floor, since a log axis cannot show zero.

Angelopoulos and Bates, 2023]. The guarantee is distribution-free and finite-sample, attractive precisely where verifier stakes are highest. Our case study is a speech-based assessment verifier; the contribution is the failure mode, not the domain.

That guarantee attaches to *whatever score is fixed in advance*. Conformal validity says nothing about whether the score measures what its user thinks it measures. Consider one record from our corpus. The verifier's recorded distribution over $\{a, b, c\}$ was $\{1.0,\ 0.0,\ 0.0\}$, giving the true option a score of $1 - 1.0 = 0.0$: total certainty. Measured properly, the same model on the same input gives $\{0.999998,\ 1.35 \times 10^{-7},\ 2.24 \times 10^{-6}\}$, a score of 2.37×10^{-6}. Neither zero was observed, and neither was the 1.0: 0.999998 rounded to five decimals *is* 1.0. One such record is a rounding curiosity. Ninety percent of them is a calibration set whose defining feature, where its mass sits, is manufactured (Figure 1).

This failure needs only three properties, and none of them is exotic: a *lossy record of a small probability* (a fixed-decimal writer that rounds it away, a token matcher blind to the spelling it arrived under, a top-k window that omits it, or any combination), *renormalization over whatever is left*, and a downstream consumer that treats the result as a probability. None looks like a defect on its own; each is a reasonable local choice, and our own first repair of the writer re-introduced the same defect at six decimals (§3). We read the source of three widely used option-scoring interfaces and one API specification (Appendix G): bounded visibility is documented, and restricted-set normalization appears in the harnesses, legitimately there, since the candidate set is enumerated by the task rather than lost, which is exactly what our case lacked. We claim nothing about prevalence.

Conformal prediction is chosen precisely for its indifference to the model's probabilities: it needs no assumption that scores are calibrated or even meaningful, only exchangeable, and that immunity is what makes this failure quiet: practitioners who adopt conformal methods *because* confidences cannot be trusted have little reason to audit them afterwards. The immunity covers calibration, not measurement.

The system. Three corpora appear below (Table 1): the artifact of §2 was found on two renderings of an 11-item work-readiness screener, and the corrected readout and cost analysis run on a 10-item civics corpus built after it, under the corrected readout from the start.

Four words carry the counting: an *item* is one question of the instrument in play; a *record* is one channel rendering of one spoken answer to one item; a *session* is one speaker's full set of answers; and the *verdict* is the session-level pass/fail the decision rule returns. Short written answers to civics questions (the Powergrading corpus; Basu et al., 2013) are spoken by synthetic voices [rany2], degraded through four channel conditions, and transcribed with faster-whisper large-v3 [SYSTRAN, Radford et al., 2023]. Two LLM judges, seated as *judge* (Qwen3-32B-AWQ; Qwen Team, 2025) and *evaluator* (Gemma-3-27B-IT; Gemma Team, 2025) and served by vLLM 0.25.1 [Kwon et al., 2023], then grade each transcript independently, each with one decoding call whose first token is

Table 1: The corpora, in the order the work used them. All speech is synthetic and degraded through telephone and noise channels. "Graded" counts bank records with a scoring key; the screener records were written by the original readout and later re-extracted under the corrected one.

Corpus	Variation	Items	Options	Graded records	Used in
Screener	channel, noise seed	11	3	660 of 770	§2–§4
Screener	accented voice	11	3	1,584 of 1,848	§2–§4
Civics (Powergrading)	speaker, channel	10	2	13,992	§3–§5

read as the decision. Option probabilities are read from that token's top-k log-probabilities and combined across judges by minimum nonconformity; a split-conformal threshold per item turns scores into per-item uncertainty sets; and a deterministic validator (the *gate*) *releases* a verdict for a session, rather than withholding it, only if every combination reachable from those sets yields the same verdict: equivalently, if the smallest and largest attainable totals fall on the same side of the mark. That rule is a counting rule: pass if at least 6 of the 10 items are correct; for a k-item subset the mark scales with k, so a single item is its own decision. Per-item budgets compose by the union bound ($\Sigma\alpha = k \cdot \alpha_i$ at a common α_i), and the exchangeability unit is the speaker; Appendix A gives the arithmetic.

The readout is lossy in three places: a fixed-resolution record, a token matcher that recognizes only some spellings, and a shared window, which §2 takes apart. The window is shared with the whole vocabulary: the constraint parameter our harness passed does not exist in the serving stack's version and is silently ignored, so the corpus was extracted unconstrained.[1] The object of study is not the pipeline but the *measurement* that enters calibration.

What conformal calibration does, and why an atom breaks it. Split conformal holds out examples whose answers are known, scores each one, sorts those scores, and reads a threshold off the sorted list, the $\lceil (n+1)(1-\alpha) \rceil$-th, where α is the error budget its user chooses. At test time an option joins the released set if its score falls under that threshold, and a verdict is released only when the set is decisive. The guarantee is real but narrow: coverage of at least $1 - \alpha$ for *whatever score was fixed in advance*, assuming nothing about whether that score is calibrated or even meaningful. That indifference is the method's selling point, and exactly what makes this failure quiet: a threshold read off a sorted list is defenceless against a pile of identical values. When ninety percent of the scores are the same manufactured 0.0, the quantile lands inside that pile and the gate stops responding to evidence, while every guarantee it advertises still holds.

Two literatures ask whether a model's confidence can be *trusted*; the conformal literature offers guarantees that hold regardless. We ask a third question: whether the number recorded as a confidence was *observed at all*, and show that when it was not, the guarantee is untouched, the judge's decisions are unchanged, and the defect surfaces only in the calibration layer. What we add is not the concept, which measurement theory and missing-data statistics have long carried, but its instance: this failure inside a distribution-free guarantee, whose own indifference to the score keeps it hidden. The remedy is correspondingly boring: measure every option or refuse.

Concretely, we contribute:

1. **An exact-zero artifact in a verification signal** (§2): on 52.3% of judge and 80.1% of evaluator extractions only one option is left with a nonzero record, so its probability becomes 1.0; under min-combination across two judges, 90.6% of graded records' true-option scores are exactly 0.0, and the calibrated threshold is exactly 0.0 from $\alpha = 0.096$ upward. Re-running the original readout on a retained window separates the causes: repairing the five-decimal writer alone removes 7,733 of them, the token matcher alone 256, the top-k window alone *none*; counts that overlap rather than partition (Table 2).
2. **Correct measurement, independently checked** (§3): a readout that measures every option's probability or refuses (absence raises; full-precision serialization), validated on 13,992 records (0 exact zeros) and, for both extractors, against a full-vocabulary reconstruction that shares neither its transport nor its window.

[1] No recorded number is affected: masking shifts surviving log-probabilities by a constant, so the renormalized distribution is invariant, and the leading token is an option on 346 of 347 audited records regardless.

3. **A pre-registered probe** (§4) separating what the artifact distorted (calibration inputs) from what it did not (argmax decisions), including the one substantive falsification and five vacuous ones at operating points where nothing releases; all ten are reported (Appendix C).
4. **What it costs** (§5): re-imposing the defect on measured scores drives the calibrated threshold to an endpoint on every item it reaches, at every floor we tried. At the writer's own 5×10^{-6}, release rises in 36 of the 42 item-splits reached and falls in none, and on the split reported, wrong verdicts rise $57 \rightarrow 59$ (aggregate error is worse in 13 of 20 resplits). At $\tau = 10^{-2}$ it instead *falls* (19 of 20), so watching it would read the defect as an improvement (Appendix F).
5. **A diagnostic another team can apply** (§7): five questions about a readout that expose the failure we demonstrate before its output reaches a calibration set, including the one our own defect got past.

We frame this at the level of the workshop's question, the *verification of the verifier's evidence*: the gate went on returning well-formed verdicts while the evidence beneath them was fabricated, with nothing in its behaviour to separate the two.

2 The artifact: asserted zeros

The two screener corpora of Table 1 predate the civics corpus: the same 77 spoken utterances under two channel-and-voice draws (Appendix E). The original readout took option probabilities from the single decision token's top-20 log-probabilities, renormalized over the options present, and wrote the result to JSON through Python's `round(p, 5)`. Three routes in that pipeline reach the same record. An option can be absent from the window; present but spelled in a way the reader's token matching does not recognize; or counted correctly and still fall below 5×10^{-6}, which the serializer writes as 0.0. What unites them is not the window: *a value that could not be represented was written as the value zero*.

Which route acted is measurable, because the readout is re-runnable: we re-extracted both corpora keeping the raw window, cut it to the top 20 by rank, aggregated under the old token matcher, renormalized, rounded to five decimals, and that reconstruction reproduces the recorded pattern of zeros on every record that has one. Two questions come apart, and only the second is causal: what did the reader have in hand when each zero was written, and what would repairing one factor alone change?

Table 2: Which factor erased each zero first, and which zeros repairing that factor alone would remove. The columns count the same 7,861 zeros and must not be added: the left one partitions them, the right one cannot, since a zero may be removable by several repairs or by none alone. "First" is pipeline order: window, then token matcher, then writer.

Factor	Zeros it erased first	Zeros that repairing it alone removes
Writer, five decimals	7,733	7,733
Token matcher, bare letters only	97	256
Window, top-20 rank cut	31	**0**

Every zero that some single repair removes is removed by repairing the writer; the other 128 need the writer *and* a second factor. Repairing the window removes not one of the 7,861 zeros: wherever an option fell outside it, the writer would have erased it anyway. The rank window is not merely secondary but entirely preempted. The token matcher is a real contributor: 97 zeros where the option sat in the window under a spelling the reader could not match, and 256 that repairing it alone removes. This artifact is a record format that could not represent a small probability, primarily by rounding it away, in a minority of cases by never counting it.

The consequence follows mechanically whichever route acted. When only one option is left with a nonzero record, its probability becomes 1.0, by renormalization where the others were never counted, by the writer where they were counted and rounded away. The record's resolution is the limit of what it can distinguish: across all 7,861 recorded zeros we re-scored, the largest true probability behind one is 1.84×10^{-5} and the median is 2.55×10^{-8}. Nothing above 10^{-4} was ever hidden, but everything below the writer's precision was, and that is exactly the range a calibration threshold reads. On

the two corpora that is $52.3\%/51.4\%$ (judge) and $80.1\%/79.5\%$ (evaluator) of all extraction rows, the two figures in each pair being the two corpora (770 and 1,848 rows).

That is the per-judge picture. The pipeline then combines two separately-prompted judges by minimum nonconformity ($s(o) = 1-p(o)$; $\min$ across judges), so either judge collapsing to a one-hot on the true option forces the combined true-option score to exactly 0.0. On the graded records this occurs on $90.6\%/89.7\%$ of rows, the spike visible in Figure 1. The combined share therefore exceeds either judge's own. The two judges are one model family per role, so their errors are correlated and we do not predict the combined rate from the marginals.

The calibration consequence is immediate: a split-conformal threshold is the $\lceil (n+1)(1-\alpha) \rceil$-th order statistic of the calibration scores. Once the zero count reaches that order statistic the threshold is exactly 0.0: at a 90.6% zero share, $\alpha \geq 0.096$; at 89.7%, $\alpha \geq 0.104$. Crossing it does not change the released set, since the five-decimal grid leaves no score between 0 and 10^{-5}: an item's set holds the options recorded at 1.0 and excludes the rest. A set that ends up empty is promoted to the full option set, withholding rather than asserting. At a threshold of 0.0 the set is therefore whichever options some judge scored 1.0: a singleton when the judges' one-hots agree, a doubleton when they disagree on a three-option item, and the full set when neither survives. The gate appears sharp; the sharpness is the artifact.

We emphasize what this is not. It is not a violation of conformal validity: the coverage guarantee of §1 holds here. What the artifact destroys is its *sharpness*. Exact coverage additionally requires a continuous score distribution; ours carries an atom of 90.6% at exactly 0.0, so realized coverage is pinned near 0.906 for every α above the crossover: conservative, and nearly insensitive to the budget its user chose. Ties are a known pathology, and smoothed conformal randomizes through them by construction [Vovk et al., 2005]; it would restore a non-degenerate threshold here without restoring any evidence, since the atom it spreads was manufactured rather than observed. And 0.906 is not a calibration property: at threshold 0 the true option is covered exactly when some judge recorded it at 1.0, so coverage is the rate at which a grader's survivor is the truth. The gate has become a detector of grader agreement. Nor is it usefully described as an implementation bug, though it began as one: everything downstream (thresholds, set sizes, release decisions) inherited a rounded, renormalized window while remaining formally correct. The defect is invisible to conformal machinery because it is upstream of it, and to ordinary testing because, as §4 shows, it changes almost nothing a system's behaviour would reveal.

3 Measuring instead of asserting

The corrected readout makes one decoding call per record and computes each option's probability as the *sum over its token variants* ("a", "A", " (a)", "a.") in the decision token's top-k (k=100), since a model spells one choice several ways and any single spelling understates it. Two hard guards define the semantics: (i) an option with no variant anywhere in the window raises an error: absence is never recorded as zero, which was the equation that produced the artifact of §2; (ii) the unnormalized option mass is persisted, so the residual carried by non-option tokens is visible rather than silently absorbed. The score is computed on the renormalized distribution; the unnormalized mass is recorded beside it and never enters the score.

The refusal rate is the natural cost of "measure or refuse", so we report it: zero. Both caches hold exactly 13,992 rows and the probe holds no unscoreable records, a check rather than an assertion, since the readout raises and nothing catches it, so one refusal would have halted the run mid-file.

Two incidents in our own deployment show how easily this defect is re-introduced, even by a team actively looking for it. First, the decoding constraint we believed we were imposing never was: our harness passed a parameter name the serving stack had retired, and unknown parameters are ignored at every arity rather than rejected, so the readout measured an unconstrained model throughout. A readout that scores each option in its own call (how we first rebuilt it) inherits that assumption and fabricates a plausible near-uniform distribution, silent exactly in the borderline band that decides calibration; an identity guard (the emitted token must equal the option asked for) turned $\sim$28,000 would-be silently wrong scores into one diagnosed defect. Second, our own first writer rounded to six decimals, returning a measured 4.3×10^{-7} as exactly 0.0 on disk: *the same defect as §2, re-manufactured by the fix itself.* A smoke test caught it; a regression test pins it now. That is no

coincidence: the writer is the dominant route into the original artifact too, and it returned the moment we stopped watching.

We then built the civics corpus (Table 1) *under the measured readout from the start*: 350 synthetic speakers $\times$ 10 items $\times$ 4 channel conditions, less two unspeakable responses disclosed in the pre-registration, graded by the same two blind judges. The *same-record* before/after lives in Figure 1 and the probe of §4; re-extracting the original corpora under the corrected readout leaves 0 of 660 (and 0 of 1,584 on the second corpus) at exactly zero, from minima of 4.1×10^{-10} and 5.2×10^{-10}. It is the substrate for §5. Under the measured readout: exactly-zero true-option scores number $0/13{,}992$; no option is recorded at exactly 1.0; per-extractor minimum probabilities are 7.8×10^{-9} and 3.9×10^{-9}; and the unnormalized option mass separates the judges: the first never records below 0.99 (minimum 0.99987), the second has a real tail (36 values below 0.9, minimum 0.4379), so where variants leave mass elsewhere, the readout now says so instead of absorbing it.

We can also report where the options sat in the window: on an audited sample of each extractor, each option's *best-spelled* variant has median rank 2 or better, never exceeds 4, and none of the 1,388 ranks falls outside a top-20 window (a statement about where an option is *findable*; most individual spellings sit far lower). For this extractor on this two-option corpus, then, a rank cut at 20 could not have produced the artifact at all: the same negative result §2 reaches on the three-option corpora, where repairing the window removes none of the zeros.

Checking the fix against something other than itself. A readout that validates only against its own output is exactly what this paper warns about, so for *both* extractors we reconstructed the same quantity from the full vocabulary, outside any server, on 347 records each (Appendix D). Four predictions were registered before the measurement; all four held on both. Option mass agrees with the deployed readout to a median ratio of 1.000000 (99th-percentile deviation 0.013 evaluator, 6.6×10^{-7} judge); scores agree to median ratios of 1.000000092 and 1.000000000, every one within a factor of two; no score is exactly zero; and the spellings the readout aggregates over carry all but 6×10^{-12} (evaluator) and 2.3×10^{-7} (judge) of the option mass.

Four limits. The reconstruction runs on the deployment's inference engine, so it validates the *readout*, not the engine or serving stacks we did not use. It shares the corrected readout's token matcher, so it cannot detect a spelling that matcher misses; that class is bounded separately, by enumerating the vocabulary for option spellings the matcher rejects. Those missed spellings carry more mass on the judge than the evaluator (2.3×10^{-7} against 6×10^{-12}, above the smallest option probabilities we report), but on no record does that mass reach the smallest option's own, at worst 13% of it. And on a deliberately adversarial stratum the evaluator's agreement loosens by an order of magnitude (mass ratios 0.90 to 1.16) while the judge's does not; we report that stratum apart rather than pooling it into the headline.

4 A pre-registered probe: what the artifact did and did not distort

Ten predictions are registered across two documents ($4 + 6$), and all ten are reported: four held, one was substantively falsified, and five were falsified *vacuously*, at registered operating points where nothing releases at all (Appendix C). Both, one for the probe below and one for the corpus of §3, were frozen before the harness that scores them existed, an ordering visible in the artifact's history and timestamped by our own record-keeping rather than a third party. The one that carries weight here is the corpus document's fourth, registered before the corpus existed: that after the fix the share of calibration scores at exactly 0.0 would fall below 0.20 (from $\sim$0.90) and the calibrated threshold would be strictly positive in every split seed. It held, its operating points amended before scoring (Appendix C).

Before re-extraction we froze four predictions about what correct measurement would change on the *original* corpora (200 records each, both judges, stratified by *recorded support*: the number of options recorded nonzero, so support-1 rows are exactly those collapsed to a one-hot). Three held; one failed; we report all four. They were not independent: Q1 is close to forced by the fix and Q4 follows from it; Q2 and Q3 were the two genuinely at risk.

Held. Q1: on every support-1 row whose surviving option was the true one, the measured score is strictly positive ($100/100$, $156/156$, $99/99$, $162/162$ across pool$\times$judge): the zero was always an artifact. Q3: argmax decisions are robust: $0/200$ flips on every stratum, which bounds the flip rate

on the probed sample at 1.5% (exact one-sided 95%; the rule of three). We do not extrapolate that bound to the full corpus: the probe was stratified by recorded support and drawn from the original corpora, so it is not a random sample of anything larger. "Unchanged" is what we observed, not what the probe guarantees. Q4: the calibration dial comes alive: the α=0.1 threshold moves from 0.0 to 1.3×10^{-3} (1.7×10^{-4} on the second corpus), both on the probe's stratified 200-record sample, an illustration of the dial rather than an operating point, with zero-score share $\sim 0.90 \rightarrow 0.000$.

Failed. Q2 predicted the measured distributions would differ substantially in total variation (median TV ≥ 0.05 on support-1 rows). Measured: median TV runs 4.2×10^{-8} to 1.1×10^{-4} across corpus, extractor and stratum: falsified by two to six orders of magnitude. The prediction mistook a missing option for missing mass: the options that went missing were small (the largest true probability behind any zero is 1.84×10^{-5}, §2), and removing small mass barely moves a distribution. This falsification is the sharpest result of the four: the artifact barely moved the distribution's *bulk*; what it destroyed was the *tail*. The practical reading is uncomfortable for downstream users: a system built on the original readout makes the same decisions and reports the same argmaxes, so nothing in its behaviour signals that its calibration layer is consuming fabricated evidence. The artifact is invisible except where it acts.

Stated plainly: the original readout was accurate to about 10^{-6} nearly everywhere, and its whole defect was writing an exact 0 where the truth was around 10^{-6}. That sounds negligible, and for the model's decisions it is, but not for calibration, for the reason §1 gives.

5 What the artifact costs at the gate

The probe says the artifact is hard to see. This section asks what it does, by re-imposing the readout's own equation on the measured scores of the binary corpus of §3: record any option below a floor τ as 0.0, renormalize, and hold the records, the split, the calibration and the decision rule fixed. This is a controlled simulation of the mechanism and is *not* pre-registered. The shape is right: a fixed-decimal writer zeroes every probability below its resolution, so the original defect *was* a probability floor, at 5×10^{-6}, and this re-imposes the same operation, at that setting and coarser ones. It does not model either other route: the token matcher drops mass rather than rounding it, and the rank route removes no zeros when repaired (Table 2) and is inert here anyway. At the writer's own 5×10^{-6} the floor already zeroes 53.1% of calibration scores and already issues wrong verdicts measurement does not ($57 \rightarrow 59$), while no item loses all its release. The figures below use $\tau = 10^{-2}$, a coarser setting that shows the mechanism's shape; Appendix F runs the ladder between them. Two harms follow, and they scale differently. The budget dial dies: wherever the floor reaches an item the threshold sits on an endpoint, at every floor we tried. And the gate issues wrong verdicts measurement withholds: two more at the writer's own floor on the split reported, five at 10^{-2}.

Where truncation acts, it drives the threshold to an endpoint. A floor rewrites only the scores beneath it, so it misses some items: on four of the ten here (calibration error 0.060–0.069; median three across the resplits below) enough of the calibration sample already sits above τ that the quantile never moves, and threshold, release set and error count are bit-identical to the measured gate. Truncation is not uniform; it reaches items whose evidence was concentrated enough for a floor to reshape.

Where it does reach, most of the calibration sample collapses onto 0 or 1 and the $(1-\alpha_i)$ quantile lands in the mass the floor created, so the threshold takes one of two values and a single comparison decides which. The threshold is 0 exactly when at least $\lceil (n+1)(1-\alpha_i) \rceil = 106$ of the 116 calibration scores are zero: equivalently, in the collapsed regime where the floor has driven every reached score to 0 or 1, when both judges miss on at most *ten* of them. The count is the invariant, not the rate: the speaker-level split leaves $n_{\text{cal}} \in \{115, 116, 117\}$ and the order statistic allows exactly ten at each, so $10/116 = 0.086$ is this split's instance of a threshold that moves slightly with n. Below that line (the *step*), the gate admits every option the readout zeroed: release rises on all four items concerned, and it rises *including on sessions the system gets wrong*. Above it the quantile falls in the mass at one, every set is full, and the gate releases nothing at all. Figure 2 shows both regimes at $\Sigma\alpha = \alpha_i = 0.10$, with the collapse falling between the items at error 0.069 and those at 0.155, consistent with a step at 0.086, though ten items cannot separate 0.086 from α_i itself, and we claim only that the step lies where the order statistic requires.

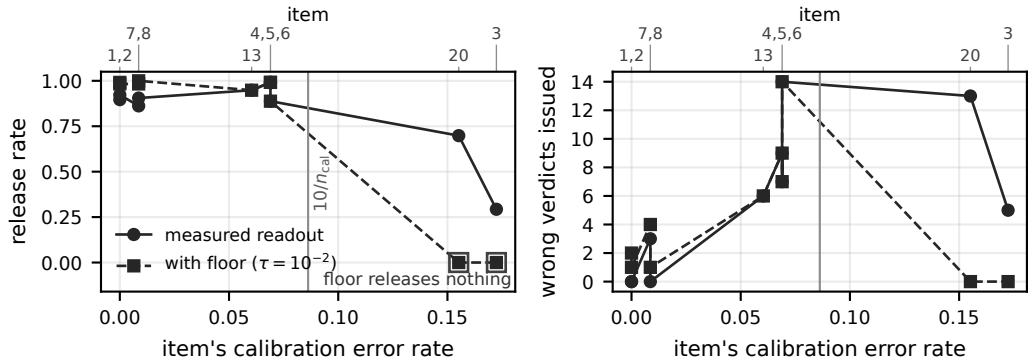


Figure 2: Ten single-item decisions at an honest budget ($\Sigma\alpha = \alpha_i = 0.10$, telephone stratum, n_{test}=116), with the floor re-imposed at $\tau = 10^{-2}$ (coarser than the five-decimal writer's own 5×10^{-6}; Appendix F runs the ladder), ordered by calibration error rate: the share of calibration records both judges score below 0.5 for the true option, a close proxy for the share the floor zeroes (1.57% of records fall between τ and 0.5; rates are channel-driven, Appendix E). The vertical line is the step, at ten wrong of n_{cal}. The floor reaches six of the ten: above the step it releases nothing (marked), and on the four below it releases more than measurement does, extra certificates that are not free (right panel). The other four keep enough calibration mass above τ that the quantile never moves, so the two readouts coincide exactly. At the writer's own 5×10^{-6} the floor changes two items, none loses its release, and wrong verdicts rise $57 \to 59$ (Appendix F, Table 3).

Certificates the measurement does not issue. Each is a single-item decision, so a release is a verdict the system issues. Below the step, truncation issues wrong verdicts that measurement does not: items 1, 2 and 8 go from 0 to 2, 1 and 1, item 7 from 3 to 4. At the writer's own 5×10^{-6} item 1 still goes from 0 to 2; it is the floor's whole cost there. This is the paper's title under a controlled floor: a guarantee that remains valid, certifying sessions on probabilities that were never observed. Above it, at this floor, the harm runs the other way: items 20 and 3 (pool IDs, not ranks) lose all 81 and all 34 of their releases, so a floor coarse enough to reach them makes the gate useless on hard items and reckless on easy ones.

When aggregate monitoring would not catch it. Summed over the ten decisions at $\tau = 10^{-2}$, wrong verdicts *fall* under truncation, 57 to 44, because the two high-error items stop releasing at all: a practitioner watching aggregate error would see improvement while per-item behaviour became arbitrary. The inversion belongs to the coarse floor: at the writer's own 5×10^{-6} aggregate error is *worse* in 13 of 20 resplits; the sign flips only once the floor silences the hard items.

What survives resampling, and what does not. These counts come from one split, at an n_{cal} where §6 puts the training-conditional band at 1.4–2.5×, so we repeated the measurement over twenty speaker-disjoint resplits of the same corpus (seeds 0–19), which bounds split sensitivity rather than replicating; a third of each split, reserved for fitting a grader, is unused, leaving n_{cal} and n_{test} near 116. At $\tau = 10^{-2}$ the threshold lands on an endpoint on every item the floor reaches ($141/141$ item-splits), at least one item collapses to zero release in $20/20$, and in $20/20$ at least one item below the step issues a wrong verdict that measurement does not. At the writer's own 5×10^{-6} the endpoint result survives ($42/42$) and so does the wrong verdict ($13/20$ splits, a different 13 from the aggregate count above, release rising in 36 of 42 item-splits and falling in none); the collapse does not.

Three things do not hold. The constants move (measured wrong verdicts 38–72 across splits, truncated 26–63, so $57 \to 44$ is an instance, not an estimate), and at 10^{-2} release below the step does not always rise: of the 117 item-splits where the floor reaches an item below the step it rises in 105, ties in 6, falls in 6, so at that floor we claim the endpoint, not a direction. Nor is the step's location a clean function of error rate: what decides it is where an item's score mass sits relative to τ (item 20 at median error 0.134 collapses in $19/20$ splits, item 3 at 0.138 in $5/20$). A fresh serving process reproduces every per-item threshold bit for bit, the inert and zero-release sets unchanged, one release count moving by one.

And at the deployed operating point, none of this is visible. There the rule is k=10 at $\Sigma\alpha \leq 0.5$, so $\alpha_i \leq 0.05$, and at α_i=0.01 the order statistic $\lceil 115.83 \rceil = 116 = n$ is the sample maximum for *any* score distribution. On the mixed-condition stream the system actually ran, nothing releases with or without the artifact, at any τ and every registered budget. That is the stream's doing, not the budget's alone: on the telephone stratum at the same operating point, 32 of 116 sessions release under measurement and none under the floor, a visible collapse.

6 Limitations and related work

Limitations. At calibration n=116 a single split's realized rates are measurements, not tests of the guarantee. Miscoverage conditional on one calibration draw is Beta-distributed in n and the order statistic alone [Vovk, 2012], with a 95th percentile at $2.5\times$, $2.0\times$ and $1.4\times$ nominal for $\alpha = 0.01$, 0.02, 0.10: twice nominal is an ordinary draw at this n, not a property of our scores, and one to quote as a band. Our corpora are synthetic-TTS speech over two assessment instruments with one model family per seat, and the constants do not carry beyond them (Appendix B records how each is checked). The mechanism (a probability recorded through a representation that cannot hold it, renormalized, then consumed as a calibration score) is not specific to speech or assessment. Appendix G cuts the other way: what we found elsewhere was not what was causal here (repairing the window removes not one zero), and the mechanism behind most of our own zeros appeared in neither system that writes a probability record. The failure is reachable, not shown to be common.

Related work. Conformal prediction and its split form are classical [Vovk et al., 2005, Angelopoulos and Bates, 2023]; training-conditional validity is due to Vovk [2012], risk control to Angelopoulos et al. [2021], and class-conditional feasibility floors to Ding et al. [2023]. Conformal methods over LLM outputs typically consume log-probabilities as scores [Kumar et al., 2023, Quach et al., 2024], credal-set predictors inherit coverage the same way [Javanmardi et al., 2024], and set-valued classification traces to Sadinle et al. [2016].

A second literature asks whether model confidences can be trusted at all: networks are overconfident [Guo et al., 2017], language models carry signal about their own correctness [Kadavath et al., 2022], verbalized confidences can beat raw logits [Tian et al., 2023], and uncertainty over generated text needs semantic equivalence [Kuhn et al., 2023]. A third studies LLM verifiers themselves and their documented biases [Zheng et al., 2023, Wang et al., 2024].

Older fields asked it first. Measurement theory separates the quantity intended from the one an instrument delivers; construct validity [Cronbach and Meehl, 1955] is the concern that a well-behaved instrument measures the wrong thing. Coding an absent value as zero is a missing-data mechanism, not at random here: an option is dropped exactly when it is small [Rubin, 1976]. The closest analogue is runtime verification, a trusted checker guarding an untrusted component [Sha, 2001].

7 A diagnostic for measurement integrity

Measurement-integrity failures are hard to test for because they leave behaviour intact: nothing in the stack raised an alarm. What can be tested is the readout, and five questions expose the one we found. *Is every option's probability present in the window you read?* A readout returning fewer options than exist is asserting the rest. *Do you keep the unnormalized mass?* Renormalizing without recording what was renormalized destroys the evidence anything went missing. *Can you name a recorded probability you never observed?* A measurement pipeline should be able to refuse; ours raises rather than write a zero it never measured. *Can the format you write to represent the smallest probability you can measure?* A window admitting every option is undone by a writer that rounds what it admits: five decimals cannot express 10^{-7}, and repairing that writer alone removes 7,733 of our 7,861 zeros. *Does the emitted token match the one you asked for?* This catches a constraint passed but silently ignored, as ours was.

All five inspect the readout, which needs its code. The distribution it produces needs only the scores: an atom like Figure 1's is visible at a glance, and is the check a consumer of someone else's harness can still run. These target the subclass we demonstrate: a value recorded without being observed, not a readout that measures the wrong conditional correctly. Conformal prediction did not fail here. A valid guarantee is only as meaningful as the measurement handed to it, and nothing inside checks the handoff.

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A The decision rule, in full

The release rule is a counting rule: a session passes if at least 6 of its 10 items are correct. For the k-item subsets used in §5 the mark scales as $\max(1, \mathrm{round}(6k/10))$, giving $1, 1, 2, 3, 6$ at $k = 1, 2, 3, 5, 10$. Per-item budgets compose by the union bound: a session is mis-set if any of its k item sets is, so $\Sigma\alpha = \sum_i \alpha_i = k \cdot \alpha_i$ at a common α_i. This needs no independence between items, which matters because ours are not independent. The exchangeability unit is the speaker: splits are speaker-level, so no speaker's records straddle calibration and test.

B The committed artifact

Every quantitative claim in this paper is recorded against the committed artifact it came from, and a build-time check re-reads the compiled PDF and requires each value printed in the content pages to trace to one of those records; the figures are produced by committed scripts that re-derive their headline shares from the artifacts on each build. Code and data are released at `https://github.com/rainsfal1/asserted-zeros`.

All model serving runs on a single 46 GB GPU with one extractor resident at a time; the two extraction passes over the binary corpus take under an hour in total. Every other result in this paper is CPU-only arithmetic over the committed score caches and re-runs in minutes, so the analysis reproduces without a GPU.

A second fabrication path exists in the same code: if *no* option survived the window, a one-hot was fabricated on the emitted token. Its rate on the original run is unmeasurable, because that run did not keep the window; re-extracting the same records with the window kept bounds the opportunity instead, and across all $5{,}236$ extractions there is none where the old reader would have found nothing to renormalize. A real defect that never fired here.

C Pre-registration bookkeeping

Of the ten registered predictions, five, all loop-policy predictions in the corpus document, were falsified *vacuously*: at their registered operating points nothing releases at all, for the reason §5 gives and not because of the loop, so we count them apart from the one substantive falsification (Q2 in §4): a prediction refuted by an inert operating point is information about the configuration, not the hypothesis. The corpus document's fourth prediction had its scored operating points amended before scoring, from $\Sigma\alpha \in \{0.05, 0.10\}$ to $\{0.10, 0.20\}$, because at the lower budgets nothing releases and the prediction would have been decided by arithmetic rather than evidence; the amendment is dated in the registered document and is additions-only.

For completeness of the registration record, a third frozen document (a feasibility-frontier study, five predictions) is registered in the artifact and is not reported in this paper. It ships with the artifact; its predictions are scored there.

D How the full-vocabulary reconstruction works

For each sampled record we take the prompt the deployed extractor sent, assert its token ids are identical to the server's own tokenization of the same messages, and run one forward pass outside any server. The next-token distribution is read at the decision position over a window of 2048 of a $151{,}669$- and $262{,}145$-token vocabulary. Option mass is the sum over all tokens that normalize to an option. Some option spellings fall outside that window on every record, so tokens outside it are *bounded* rather than assumed zero: each contributes at most the window's smallest probability, giving a relative bound of 1.45×10^{-13} (evaluator) and 5.3×10^{-11} (judge). The comparator is a fresh extraction from the same server instance rather than the committed cache, because a long-running server does not reproduce its own earlier outputs exactly: 154 of 200 records bit-identical against the day-old instance that wrote the cache, worst deviation 1.27×10^{-2}. A clean relaunch, by contrast, reproduces the cache almost perfectly (99.98% and 99.62% of $13{,}992$ rows bit-identical per extractor), so the instability is within-lifetime state, not launch nondeterminism; the same-instance comparator guards against both.

E The corpora of §2

Both corpora serve an 11-item spoken work-readiness screening instrument (options a/b/c per item; the scoring key is transcribed from a deployed system and pinned in the artifact). Utterances are rendered by synthetic voices [rany2], degraded, and transcribed with faster-whisper large-v3 [SYSTRAN]. The first corpus: 77 utterances (66 answer-bank, 11 clean probes) $\times$ 5 channel levels (telephone band-limiting and additive noise at signal-to-noise ratios $+5/0/-5/-10$ dB) $\times$ 2 noise seeds $=$ 770 rows, of which 660 are graded bank records. The second: the same 77 utterances $\times$ 8 accented voices $\times$ 3 levels (telephone; SNR $0/-5$) $=$ 1,848 rows, 1,584 bank. The binary civics corpus of §3 uses 4 channel draws (telephone; SNR $0/-5/-10$). In §5, the per-item error rates that place an item above or below the step are themselves channel-driven: the share of calibration records both judges score below 0.5 runs 5.4% to 42.9% across our four conditions, tracking word-error rate (1.15 on those records against 0.094 elsewhere), so the recording, not the grader alone, sets how much error a gate must tolerate.

F The τ-ladder

§5's figures use $\tau = 10^{-2}$. The same measurement at the defect's own scale, on the same stream and split:

Table 3: §5's per-item measurement repeated at four floors, including 5×10^{-6}, the writer's own resolution.

τ	Cal. scores at 0.0	Gates changed	Zero-release	Wrong
0 (measured)	0%	—	0	57
10^{-6}	46.3%	1	0	57
5×10^{-6} (the writer's)	53.1%	2	0	**59**
10^{-4}	75.8%	3	0	60
10^{-2} (§5)	91.3%	6	2	44

At the writer's own resolution the floor already zeroes half the calibration mass and the gate already issues wrong verdicts that measurement does not ($57 \to 59$); the zero-release collapse, and with it the fall in aggregate error that makes the defect read as an improvement, appear as the floor coarsens. Repeating §5's twenty resplits at each rung gives the same separation: endpoint thresholds at $27/27$, $42/42$, $58/58$ and $141/141$; no zero-release item below 10^{-2}; and aggregate error worse under the floor in 0, 13, 15 and 1 of 20 splits as τ coarsens. Post-hoc, as §5 states.

G A lightweight audit of related interfaces

To ask whether the assumptions our failure needs are peculiar to us, we read the source of three widely used option-scoring interfaces and one API specification on 2026-08-19. All four are moving targets and we did not record the commits we read, so the reading is dated rather than pinned: a reader cloning later may find any cell changed. The table records assumptions, not defects. Restricting to an enumerated candidate set and normalizing over it is correct where the measured quantity is explicitly conditional; that is ordinary scorer behaviour and is not the failure this paper documents. We make no claim about prevalence, and none about the correctness of any system below.

Table 4: Four option-scoring interfaces, each read at the level its Evidence column states: [V] we read the implementation, [D] we read only documentation or a type declaration. A dash means the property was not found on the path we read.

System	Evidence	Precision	Candidate visibility	Renormal.	Interpretation
OpenAI chat API	spec text [D]	—	capped at 20	—	bounded visibility is documented in the specification at 20, the same width we read
`lm-eval-harness`	source [V]	full float in the record	exact (echo); raises on the chat path	over the task's choices	the restricted-set assumption exists, and is legitimate here: the choices are enumerated, so the value is conditional by construction
`lighteval`	source [V]	full float in the record	—	over the task's choices	likewise legitimate: the candidate set is defined by the task, not lost; no precision loss on the serialized record
`inspect_ai`	config [D], path [V]	—	0–20 declared	none	the visibility limit exists in configuration; its multiple-choice path parses text and never consumes probabilities

Two results are worth separating. Bounded visibility and restricted-set normalization each appear outside our system, so the assumptions are not ours alone, though wherever we found the normalization, it was over a candidate set the task enumerates, which is precisely what our case lacked. But the route that produced most of our own zeros, a writer that cannot represent a small probability, appears in *neither* harness that writes a probability record: `lm-eval-harness` and `lighteval` serialize in full float, and `inspect_ai`'s multiple-choice path writes none. And in one case the discipline §3 recommends is already practised: on its chat-completions path `lm-eval-harness` raises rather than returning an approximation, giving the unavailability of prompt log-probabilities as its reason. We report that as behaviour only; the source says nothing about calibration, so we claim no shared motivation.

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   Justification: §5's headline counts come from a single deterministic speaker-level split, so the paper repeats the entire measurement over twenty speaker-disjoint resplits and reports the spread rather than error bars: which claims hold across all twenty ($141/141$ endpoint thresholds; $20/20$ zero-release; $20/20$ wrong certificates below the step), and which do not (wrong-verdict totals ranging 38–72 measured and 26–63 truncated; the aggregate drop reversing at one seed; release below the step falling rather than rising in 6 of 117 item-splits). §6 gives the reason single-split rates cannot be read as tests of the guarantee (at this n the training-conditional band runs 1.4–$2.5\times$ nominal, Vovk, 2012), and the probe's one interval claim carries its exact bound ($0/200$ argmax flips, one-sided 95% upper bound 1.5%).

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